



# Comparative evaluation of ChatGPT-4, ChatGPT-3.5 and Google Gemini on PCOS assessment and management based on recommendations from the 2023 guideline

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Received: 25 July 2024 / Accepted: 23 November 2024 / Published online: 2 December 2024  
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## Abstract

**Context** Artificial intelligence (AI) is increasingly utilized in healthcare, with models like ChatGPT and Google Gemini gaining global popularity. Polycystic ovary syndrome (PCOS) is a prevalent condition that requires both lifestyle modifications and medical treatment, highlighting the critical need for effective patient education. This study compares the responses of ChatGPT-4, ChatGPT-3.5 and Gemini to PCOS-related questions using the latest guideline. Evaluating AI's integration into patient education necessitates assessing response quality, reliability, readability and effectiveness in managing PCOS.

**Purpose** To evaluate the accuracy, quality, readability and tendency to hallucinate of ChatGPT-4, ChatGPT-3.5 and Gemini's responses to questions about PCOS, its assessment and management based on recommendations from the current international PCOS guideline.

**Methods** This cross-sectional study assessed ChatGPT-4, ChatGPT-3.5, and Gemini's responses to PCOS-related questions created by endocrinologists using the latest guidelines and common patient queries. Experts evaluated the responses for accuracy, quality and tendency to hallucinate using Likert scales, while readability was analyzed using standard formulas.

**Results** ChatGPT-4 and ChatGPT-3.5 attained higher scores in accuracy and quality compared to Gemini ( $p = 0.001$ ,  $p < 0.001$  and  $p = 0.007$ ,  $p < 0.001$  respectively). However, Gemini obtained a higher readability score compared to the other chatbots ( $p < 0.001$ ). There was a significant difference between the tendency to hallucinate scores, which were due to the lower scores in Gemini ( $p = 0.003$ ).

**Conclusion** The high accuracy and quality of responses provided by ChatGPT-4 and 3.5 to questions about PCOS suggest that they could be supportive in clinical practice. Future technological advancements may facilitate the use of artificial intelligence in both educating patients with PCOS and supporting the management of the disorder.

**Keywords** artificial intelligence · chatGPT · gemini · polycystic ovary syndrome · ovulatory dysfunction · infertility

## Introduction

Polycystic ovary syndrome (PCOS) is a prevalent condition affecting up to one in five women of reproductive age [1]. The syndrome is characterized by ovulatory dysfunction, hyperandrogenism and polycystic ovarian morphology

[2, 3]. PCOS can lead to a variety of complications, including infertility, metabolic disorders and psychological challenges, underscoring the importance of early diagnosis and comprehensive management [2]. Developing a comprehensive plan is crucial for the treatment of PCOS. A critical component of this plan is patient education, which should be derived from credible sources and encompass both the short and long-term consequences of the condition. This education is essential for patients to gain a comprehensive understanding of their diagnosis, alleviate anxiety, and take proactive measures to effectively manage and mitigate the impact of PCOS throughout their lives [4].

The complications associated with PCOS and their profound impact on women's lives often drive patients to seek

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information from online sources to manage their condition and make necessary lifestyle changes [5]. This is particularly relevant in countries where healthcare services may be inadequate in addressing their concerns. Heightened concerns about body image among PCOS patients often lead them to seek answers and support online rather than consulting healthcare professionals, underscoring the need for accessible and reliable digital health resources [6].

Over the past few years, use of artificial intelligence (AI), especially advanced language models like ChatGPT has become increasingly prevalent in the medical field, offering valuable support to both patients seeking information and healthcare professionals looking for quick, reliable insights [7]. ChatGPT is an AI language model created by OpenAI, utilizing the Generative Pre-trained Transformer (GPT) architecture. This system, similar to other large language model-based AIs, has been trained on extensive datasets of text, enabling it to generate new text that mirrors the style and content of its training material [8]. This capability involves not just generating text, but also understanding and interpreting human language through advanced computer algorithms [9].

ChatGPT-3.5 was released by open AI for development purposes and limited testing in March 2022, followed by its official commercial launch on November 22, 2022. During this time, efforts were focused on ensuring the model's reliability, effectiveness for the general public and expanding access to a broader audience. The latest update for ChatGPT 3.5 was released on August 3, 2023, aimed to improve the user experience by adding features like suggested replies and ensuring that user settings are more stable [10, 11]. Following its release, ChatGPT-3.5 was quickly adopted in the medical field to support patient education, clinical decision support, and various healthcare tasks [12]. ChatGPT-4 boasts, which is released in March 2023, a higher parameter count and enhanced computational power compared to its predecessor, ChatGPT-3.5. The latest update of ChatGPT-4 was released on July 20, 2023 focusing on mathematical capabilities and logical reasoning. This update also brings improvements in language capabilities, speed and the ability to manage more complex conversations [10, 13]. The free version of ChatGPT powered by GPT-3.5 model shows significant utility in informal contexts, provides basic features with slower speeds and limited access during busy period. The paid version uses the GPT-4 model, offering higher rate limits and producing higher-quality and more accurate responses [14]. Consequently, ChatGPT-4 is better equipped to handle more complex queries and tasks with greater efficiency and accuracy [15]. Additionally, it began being utilized in the medical field shortly after its release [16].

Google's most advanced artificial intelligence model, Gemini, was released in December 2023. It was formerly

known as Google Bard before being rebranded and enhanced to handle more complex, multimodal tasks [17]. The free version of Google Gemini received its last update on July 26, 2024. This update included the enabled file uploads, improved language support and added features to enhance response accuracy and also the introduction of the Gemini 1.5 Flash model [18]. Increasingly used in medicine today, Gemini is a powerful multimodal AI that elevates diagnostic accuracy, improves personalized treatment plans, and provides real-time decision support by integrating and analyzing images, audio, video and text [19].

The growing reliance on large language models for medical information indicates that PCOS patients may increasingly depend on these technologies due to their easy accessibility and prompt assistance [5]. This trend highlights the critical need to provide accurate and supportive guidance for managing PCOS through these platforms. However, there is a notable absence of studies in the literature that evaluate the performance of large language models in providing information on PCOS management and treatment. Moreover, no research has yet compared the effectiveness of these models.

This study aimed to evaluate the accuracy, quality, tendency to hallucinate and readability of responses from leading large language models—ChatGPT 3.5, ChatGPT 4 and Gemini—by comparing their answers to questions about PCOS against current clinical guideline and also against each other. The primary reason for utilizing ChatGPT and Gemini in this study is their widespread application in the medical field [20]. Both models have been the subject of several studies regarding their use in medical contexts. Using these tools with existing body of research and data allows for a more comprehensive analysis [7, 12]. Other large language models with still limited use in medical context were not included. The latest PCOS guideline, titled 'Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome,' was used as a benchmark in the current study [21]. This guideline, updated in 2023, offers up-to-date and comprehensive recommendations for assessing and managing PCOS, ensuring our research aligns with the latest standards.

## Materials and methods

### Study Design

This is a cross-sectional study evaluating AI performance in providing information on PCOS by comparing responses to current clinical guideline, assessing their potential medical usability. The three most popular chatbots were used in this

study: ChatGPT-4, the paid version; ChatGPT-3.5, the free version; and Gemini, the free version.

The questions were developed by two endocrinologists using the most recent PCOS guideline, ‘Recommendations from the 2023 International Evidence-based Guideline for the Assessment and Management of Polycystic Ovary Syndrome.’ Additionally, in the final section, frequently asked questions from PCOS patients in our endocrinology clinic were collected.

A total of 44 questions were included in the study, divided into four groups: Diagnosis (13 questions), long-term health risks (8 questions), treatment options (12 questions), and frequently asked questions by individuals with PCOS (11 questions) (Table 1).

This research did not involve human or animal participants, thereby eliminating the requirement need for ethical consent and approval.

### Analysis of responses from AI models

Each question was tested by one endocrinologist and one internist using all three chatbots—ChatGPT-4, ChatGPT-3.5 and Gemini between June 8–9 2024. Raters were kept unaware of the chatbots’ identities ensuring the maintenance of objectivity in the assessment process. The responses from ChatGPT-4, ChatGPT-3.5 and Gemini were independently evaluated by graders using the current PCOS guideline as a benchmark, assessing the accuracy, quality, tendency to hallucinate and readability of the responses. Graders scored the answers independently for accuracy, tendency to hallucinate and quality. Responses with misleading statements were regarded as inaccurate. Finally, a combined score was created from the scores given by the graders.

The accuracy of the chatbot’s answers was evaluated by the graders using a 6-point Likert scale, ranging from 1 (completely incorrect) to 6 (completely correct), by comparing them with the current PCOS guideline [21]. These responses categorized into three distinct levels: poor accuracy (1–2 points), borderline accuracy (3–4 points) and good accuracy (5–6 points). The quality of chatbots’ responses was evaluated using a 5-point Likert scale, ranging from 1 (poor quality) to 5 (excellent quality). Then these responses categorized into three distinct levels: low quality (1–2 points), moderate quality (3 points), and high quality (4–5 points) [22]. The tendency of hallucinate of responses was evaluated by graders using a 5-point Likert scale, ranging from 1 (completely incorrect) to 5 (completely correct).

Readability was analyzed using Readable software (Readable.com, Horsham, United Kingdom) [23]. The readability of the answers was evaluated using three widely recognized metrics: the Flesch Reading Ease (FRE) score, the Flesch-Kincaid Grade Level (FKGL), and the Gunning Fog Index (GFI).

### Data analysis

The Kolmogorov-Smirnov test was employed to assess the normality of the distribution of the data. Numerical variables were represented by mean or median values, while categorical variables were expressed as percentages. The accuracy and quality of the AI models were assessed using frequencies and percentages and compared via Chi-square tests. In the comparisons made between chatbots, the mean values of accuracy, quality scores, and readability values were calculated and compared using the one-way ANOVA method. All statistical analyses were performed using SPSS version 26.0 (IBM Inc., Armonk, NY, USA). A *p*-value of less than 0.05 was set as the threshold for statistical significance.

### Results

The highest accuracies were observed as 68.2%, 56.8%, and 29.6% for ChatGPT-4, ChatGPT-3.5, and Gemini, respectively (*p* = 0.002). ChatGPT-4 showed similar accuracy to ChatGPT-3.5 (*p* = 0.52) but significantly outperformed Gemini (*p* = 0.001). Additionally, ChatGPT-3.5 demonstrated higher accuracy compared to Gemini (*p* = 0.007).

The highest reported quality scores were 75% for ChatGPT-4, 59.1% for ChatGPT-3.5, and 29.5% for Gemini (*p* < 0.001). ChatGPT-4 showed comparable accuracy to ChatGPT-3.5 (*p* = 0.1) but significantly outperformed Gemini (*p* < 0.001), whereas ChatGPT-3.5 also demonstrated higher accuracy compared to Gemini (*p* < 0.001). There was a significant difference between the tendency to hallucinate scores, which were due to the lower scores in Gemini. (*p* = 0.003). There were no significant differences between ChatGPT-3.5 and 4 (*p* = 0.46). Comparison between AI models are illustrated in Fig. 1.

In the evaluation of readability scores, ChatGPT 3.5 and ChatGPT 4 had similar scores, while Gemini demonstrated higher readability (*p* < 0.001) (Table 2).

When evaluating between question groups, it was observed that the accuracy, quality and tendency to hallucinate scores for the answers given to the frequently asked questions group were lower across all chatbots. The difference in accuracy and quality scores between ChatGPT-3.5, ChatGPT-4, and Gemini was largely due to this section.

Analysis of ChatGPT-4’s responses based on different question categories were shown in Table 3. Tendency to hallucinate score was also found to be low in the treatment section. (*p* < 0.001) Examples of questions with the high hallucinations scores can be seen in Supplementary Figs. 1–6.

**Table 1** The questions posed to AI models**A- Diagnosis**

What is an ‘irregular’ menstrual cycle?  
 What are the current diagnostic criteria for PCOS?  
 What are the criteria for biochemical hyperandrogenism?  
 Which method should be used for androgen measurement in patients with PCOS for the most accurate results?  
 What is the clinical evidence of hyperandrogenism?  
 How is the modified Ferriman Gallwey score calculated?  
 Is it necessary to perform ultrasonographic imaging when diagnosing PCOS?  
 What is the diagnostic criteria for polycystic ovarian morphology?  
 What are the criteria for PCOM in adolescents?  
 How can we understand that menstrual cycles are anovulatory in patients with regular cycles?  
 Can AMH be used as a diagnostic criterion in the diagnosis of PCOS?  
 What are the factors affecting serum AMH levels?  
 What diseases should be ruled out when diagnosing PCOS?

**B -Long Term Health Risks**

What are the common disorders associated with PCOS in the long-term follow-up?  
 What cardiovascular disease risk factors should be monitored in PCOS patients?  
 What is the most accurate test to assess the glycemic status in PCOS patients?  
 What is the major reason of increased endometrial cancer risk in PCOS patients?  
 What are the mechanism of PCOS causing infertility?  
 What psychiatric disorders are commonly seen in patients with PCOS?  
 Is there an increased risk of metabolic disease in relatives of PCOS patients?  
 What are the common seen pregnancy complications in women diagnosed with PCOS?

**C- Treatment Options**

What are the lifestyle change strategies can be used to manage PCOS?  
 Is there a specific diet recommended for women with PCOS?  
 What exercise recommendations should be given to women diagnosed with PCOS?  
 What are the pharmacological treatment options for PCOS?  
 What is the role of metformin therapy in the management of PCOS?  
 How should be the estrogen content of COCs used in the treatment of PCOS?  
 What are the anti-obesity treatment options used in the management of PCOS?  
 What are the anti-androgen treatment options used in the management of PCOS?  
 Does laser hair removal have a place in the treatment of hirsutism in PCOS patients?  
 What is the available therapies to improve reproductive and obstetric outcomes in women with PCOS?  
 What are the treatment strategies for ovulation induction in infertile PCOS patients?  
 Does inositol have a place in the treatment of PCOS?

**D-Frequently Asked Questions by individuals with PCOS**

Does PCOS symptoms get better as you get older?  
 Can PCOS be treated with lifestyle changes?  
 Do PCOS symptoms improve after weight loss?  
 Is PCOS genetically inherited?  
 Can PCOS be cured with any treatment?  
 How long should oral contraceptives be used in the treatment of PCOS?  
 Does PCOS get cured after pregnancy?  
 Which medical specialties should a patient diagnosed with PCOS consult?  
 Which herbal treatments can be used for PCOS treatment?  
 Can onion remedy be used in the treatment of PCOS?  
 Do the cysts in the ovaries of PCOS patients disappear or become cancerous?

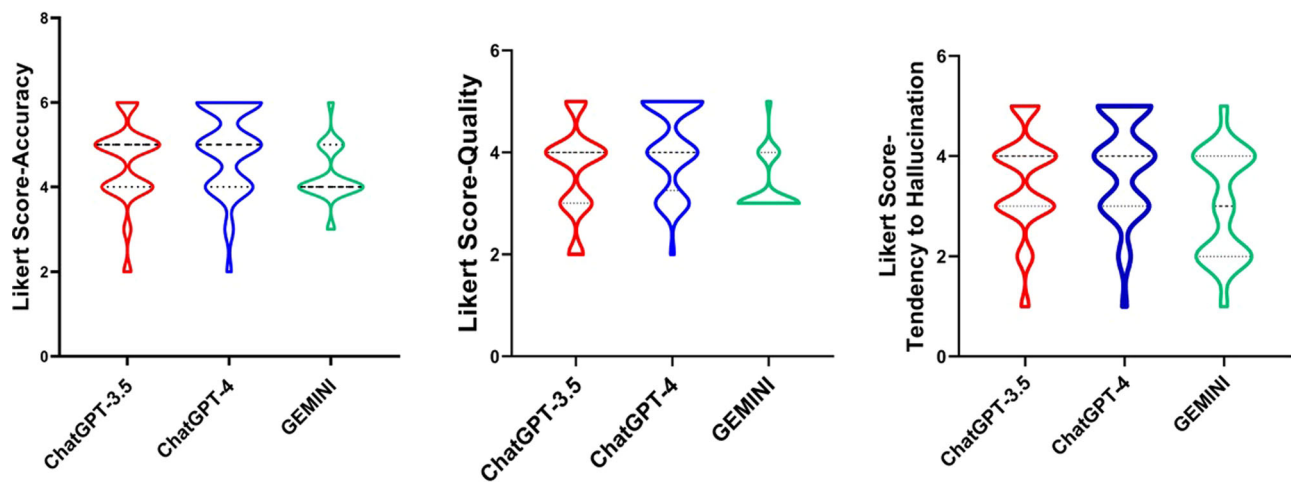


Fig. 1 Comparison between AI models

**Table 2** Analysis of Accuracy, Quality and Readability among AI models

| AI models                                | ChatGPT 3.5 | ChatGPT4    | Gemini      | P value |
|------------------------------------------|-------------|-------------|-------------|---------|
| Accuracy (Likert score)                  | 4.6 ± 0.9   | 4.9 ± 1     | 4.3 ± 0.7   | 0.003   |
| Quality (Likert score)                   | 3.6 ± 0.9   | 4.1 ± 0.9   | 3.3 ± 0.6   | 0.005   |
| Tendency to Hallucination (Likert score) | 3.5 ± 1     | 3.8 ± 1     | 3.1 ± 1.1   | 0.003   |
| Readability                              |             |             |             |         |
| FKGL                                     | 15.7 ± 2.3  | 15 ± 2.1    | 11.7 ± 1.8  | <0.001  |
| GFI                                      | 18.2 ± 2.7  | 17.5 ± 2.5  | 14.1 ± 2.1  | <0.001  |
| FRE                                      | 17.5 ± 13.3 | 21.5 ± 11.2 | 36.7 ± 10.2 | <0.001  |

AI Artificial Intelligence, FKGL Flesch-Kincaid Grade Level, FRE Flesch Reading Ease, GFI Gunning Fog Index

Data are given as mean (SD) and median (IQR) where appropriate

## Discussion

In this study, the competence of AI in the assessment and management of PCOS was evaluated by comparing the most popular chatbots in medicine. The study results showed that ChatGPT-4 and 3.5 are both quite strong and similar in terms of quality and accuracy. Conversely, Gemini was found to have higher readability compared to the others, but its quality and accuracy scores were lower. Also, Gemini was more likely to cause hallucinations compared to ChatGPT-3.5 and ChatGPT-4. These results indicate a promising potential for AI to play a role in the management of PCOS.

A broad range of large language models have been developed with the rapid technological advancements. Researchers can access valuable online resources where large language models are continually evaluated from

various perspectives [24]. ChatGPT and Gemini were chosen due to their extensive use in medical contexts, making them well-suited to this research [20]. There are some notable differences between ChatGPT and Gemini. The main difference is that the free version of ChatGPT lacks up-to-date information which means the information provided by free version of ChatGPT is limited to its last update. In contrast, the free version of Gemini is designed to get information from dynamic sources by accessing internet. Considering this difference in working methods, Gemini has been regarded as superior to the free version of ChatGPT [25]. The paid version of the ChatGPT also offers real-time information from the web and may include early access to new features and updates [26]. An important gap in the literature is that there is no information about which of these large language models is better equipped for a specific condition such as PCOS.

Numerous investigations have already begun assessing the effectiveness and reliability of ChatGPT in various areas of endocrinology. Studies, particularly those comparing current guidelines with AI, have demonstrated whether AI can keep healthcare professionals updated with the latest information [27–30]. A study comparing ChatGPT-4's performance in thyroid nodule management to the European Thyroid Association's 2023 Guidelines found notable improvements and consistency in handling multiple-choice queries. However, ChatGPT's performance in binary queries has been inconsistent, underscoring the need for further refinement in specific areas [29]. Another study comparing Bard and ChatGPT's performance on diabetes and diabetes technology showed that ChatGPT offered more detailed and comprehensive explanations, while Bard provided mostly accurate but sometimes less thorough responses [30]. A recent study assessed ChatGPT's performance in healthcare decisions for patients with diabetes and obesity, focusing on its



**Table 3** Analysis of ChatGPT-4's Responses Based on Different Question Categories

| ChatGPT-4 Responses                         | 1) Diagnosis Section<br>N = 13 | 2) Long term Health Risk<br>Section N = 8 | 3) Treatment Section<br>N = 12 | 4) Frequently Asked<br>Questions Section N = 11 | P value |
|---------------------------------------------|--------------------------------|-------------------------------------------|--------------------------------|-------------------------------------------------|---------|
| Accuracy (Likert score)                     | 4.9 ± 1.3                      | 5.5 ± 0.5                                 | 5.2 ± 1.0                      | 4.3 ± 0.5                                       | 0.04    |
| Quality (Likert score)                      | 4.4 ± 0.9                      | 4.5 ± 0.5                                 | 4.3 ± 0.9                      | 3.5 ± 0.5                                       | 0.02    |
| Tendency to Hallucination<br>(Likert score) | 4.5 ± 0.9                      | 4.5 ± 0.5                                 | 3.4 ± 0.8                      | 3 ± 0.9                                         | <0.001  |
| Readability                                 |                                |                                           |                                |                                                 |         |
| FKGL                                        | 15.1 ± 2.7                     | 16.3 ± 1.9                                | 14.3 ± 1.8                     | 14.9 ± 1.7                                      | 0.23    |
| GFI                                         | 17.2 ± 2.7                     | 19.2 ± 2.6                                | 16.5 ± 2.2                     | 17.6 ± 2.1                                      | 0.18    |
| FRE                                         | 19.8 ± 13.6                    | 14.9 ± 11.1                               | 24.7 ± 9.5                     | 24.7 ± 8.3                                      | 0.17    |

*FKGL* Flesch-Kincaid Grade Level, *FRE* Flesch Reading Ease, *GFI* Gunning Fog Index

Data are given as mean (SD) and median (IQR) where appropriate

reliability and effectiveness. The study found that ChatGPT fully adhered with guidelines for obesity assessment in type 2 diabetes but was less effective in obesity management [27].

PCOS is a prevalent condition, but studies on AI and PCOS are limited. The accuracy and reliability of ChatGPT-4 in responding to questions about PCOS and infertility were assessed in a study. Results showed ChatGPT-4 generated accurate and reliable responses, but evaluations were subjective, and dynamic learning was lacking [17]. The study's findings which demonstrates especially ChatGPT-4's potential to enhance decision-making support for healthcare professionals in managing PCOS-related infertility is encouraging. Our study found no significant accuracy and quality differences between ChatGPT-4 and ChatGPT-3.5, though both outperformed Gemini. The lack of distinction was probably caused by small number of questions. When viewed as a ratio, it was observed that evaluators gave proportionally higher scores to ChatGPT-4.

The chatbots showed gaps in their knowledge concerning certain specifics of the 2023 PCOS guideline. None of the chatbots correctly stated that there are no specific criteria for diagnosing PCOM in adolescents and that ultrasonography is not recommended for diagnosis in this age group. ChatGPT-4 and 3.5 correctly explained how to diagnose anovulatory cycles in individuals with regular menstrual cycles, while Gemini failed to mention the importance of checking progesterone levels during the luteal phase of the cycle. Additionally, ChatGPT-4 and Gemini provided updated information on inositol use in PCOS treatment, but ChatGPT-3.5 lacked this knowledge. Lastly, ChatGPT-4 offered detailed information about the estrogen content of oral contraceptives in PCOS therapy, while the other chatbots fell short.

In this study, the readability assessment revealed that the answers provided by Gemini had higher readability scores compared to those from ChatGPT-4 and ChatGPT-3.5. When evaluated as a whole, it was observed that the

readability level of all chatbots is relatively low. According to the scores, only individuals with a college education or higher could comprehend the responses effectively. In a previous study, it was also found that ChatGPT-4's responses about hypothyroidism during pregnancy found low readability, highlighting it as a key limiting factor [31]. Given that these chatbots are intended for patient education, this poses a significant limitation.

The chatbots were found to provide more accurate and detailed answers to questions generated from the guideline which is in the first three question groups. However, their responses to open-ended questions, which required answers sourced from the literature and were not directly addressed in the guideline, were quite inadequate. When evaluating the data across different question groups, the frequently asked questions group had the lowest averages in terms of quality and accuracy. This suggests that while chatbots can effectively handle structured guideline-based queries, they struggle with more complex, literature-dependent questions, highlighting an area for potential improvement. In prior studies evaluating AI's response capabilities, it has been observed that the chatbots tends to provide less accurate answers to questions with open-ended answers [32]. OpenAI, the developer behind ChatGPT, has also acknowledged on its official website that ChatGPT faces challenges when responding to questions that require interpretation [33, 34].

The increasing use of AI in the scientific community has led to concerns about the potential for generating misleading information among experts. It is important to be aware of the possibility of AI creating incorrect diagnosis or treatment recommendations despite the numerous benefits it offers [35]. A critical limitation of these models is inability to evaluate the clinical picture of a patient which has led to clinical safety concerns [36]. For instance, when queried about treatment options for PCOS, all of the chatbots recommended oral contraceptive pills. Nonetheless, the optimal treatment plan should be tailored by the

patient's physician, who has access to the patient's medical history, laboratory results, and potential side effects [37]. Another example is that ChatGPT-4 suggested the use of an onion remedy as a supplementary treatment for PCOS, despite clinical studies indicating that such remedies have no established role in the management of PCOS [38].

The results of the current study indicate that Gemini is more prone to generating hallucinations compared to ChatGPT-3.5 and ChatGPT-4. This observation aligns with a previous meta-analysis that reviewed 11 systematic reviews across various fields, which also reported higher rates of hallucinations associated with Gemini relative to ChatGPT-3.5 and ChatGPT-4 [35]. The tendency of a model to produce hallucinations is a critical factor to consider prior to its application in the medical domain. The occurrence of hallucinations in AI models highlights the essential role of healthcare professionals in making informed clinical decisions.

While this study is the first to demonstrate the potential of AI in effectively answering questions about PCOS, it does have several limitations. The subjective nature of the evaluations can be considered as a limiting factor. The lack of significant differences in accuracy and quality between ChatGPT-4 and 3.5 might be due to relatively small number of questions. Another limitation is the inclusion of only two large language models, ChatGPT and Gemini, from the many available options such as Perplexity.ai, Claude by Anthropic, Microsoft Copilot, Llama2 by Meta, and Mistral.ai that might offer better performance with general AI tasks yet with limited use in the medical context. Even though we have administered identical questions to the models simultaneously in this study, one should take into account that variability in the outputs at different times, potentially due to updates in the training datasets of the large language models, also represents a limitation in general use. Finally, the rapid evolution of AI technology suggests that the performance of these chatbots could evolve over time, potentially impacting the relevance of the study's findings.

Artificial intelligence is developing in medicine and holds promise for patient education. This is the first study to compare responses about assessment and management of PCOS from ChatGPT-4, ChatGPT-3.5, and Gemini. In terms of quality and accuracy, ChatGPT-4 and ChatGPT-3.5 outperformed, while Gemini demonstrated outstanding readability. However, all chatbots require college-level understanding, indicating a need for improvement. Improving AI capabilities is essential for reliable use in medical sciences, disease management, and patient education.

## Data availability

Data is provided within the manuscript.

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1007/s12020-024-04121-7>.

**Author contributions** B.O.Y. was the corresponding author and formulated the research question. I.G. and J.F. were responsible for formulating the questions, design, article screening. I.G. and S.A. evaluated and compared the responses from various perspectives, data acquisition and interpretation. I.G. created the first draft of the manuscript and all authors have read, reviewed and approved the final manuscript.

**Funding** This research received no specific grant from any funding agency, commercial or not-for-profit sectors.

## Compliance with ethical standards

**Conflict of interest** The authors declare no competing interests.

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